

WHY IS LIDAR ADOPTION SLOWER Than Expected?

A decade of hype. Billions invested. But lidar still isn't mainstream. What went wrong—and what does it mean for the future of autonomous vehicles?



DR XIAOXI HE,
Research Director at
IDTechEx, specialises
in technology
scouting, market
analysis, and
business insights
of autonomous
vehicles, helping
clients assess
capabilities and
identify commercial
opportunities.



Heralded as a breakthrough for the automotive industry's shift towards safer, fully autonomous vehicles, lidar technology promises a future with zero road fatalities and enhanced accessibility. Despite a decade of extensive development, testing, and financial investments such as partnerships and acquisitions, lidar-equipped vehicles remain scarce on the roads. Known for its high-resolution 3D mapping and precise depth perception, lidar is essential for driverless cars but faces significant adoption challenges.

Although lidar delivers accurate data for autonomous navigation, several obstacles hinder its mainstream integration. High costs, technical limitations in various environments, complex data management, and competition from other sensor technologies have impeded its widespread use.

Despite its advantages in depth perception and detailed mapping, lidar confronts economic and technical barriers. Performance issues in adverse weather and the substantial investment required for deployment have made manufacturers and consumers cautious, slowing the technology's adoption in the automotive sector.

The cost barrier to lidar adoption

Lidar technology, known for its precise point cloud mapping and high-resolution data, faces a significant adoption barrier due to its high cost. Lidar systems are substantially more expensive to produce and integrate than more economical sensors like cameras and radar. This cost disparity has kept lidar from becoming a mainstream option in the automotive industry, leading manufacturers to favour cheaper alternatives.